Causeway BY NETCORENETWORK

Causeway is a lightweight control panel and embeddable software development kit for establishing secure, direct connections between devices. It lets you add and remove peers, manage private connections, and reach services running on another machine as though they were available locally.

This guide will walk you through downloading, installing, and running Causeway on macOS, Windows, or Linux.

1. Download and extract Causeway

Download the Causeway version that matches your operating system and device architecture. Once the download is complete, extract the downloaded archive.

Your Causeway folder should contain:

- **cwp2p** (or **cwp2p.exe** on Windows): The Causeway executable.
- **README.md:** Setup instructions and additional information.

Keep these files together in the same folder.

2. Open a terminal in your Causeway folder

You'll use a terminal to start and interact with Causeway.

macOS

1. Open Terminal.
2. Type cd followed by a space.
3. Drag your extracted Causeway folder from Finder into the Terminal window.
4. Press Enter.

For example: cd ~/Downloads/causeway

Windows

1. Open File Explorer and navigate to your extracted Causeway folder.
2. Click the address bar, type powershell, and press Enter.
3. A PowerShell window will open in that folder.

Linux

1. Open your file manager and navigate to your extracted Causeway folder.
2. Right-click inside the folder and select **Open in Terminal**, if available. *Alternatively, open a terminal and use cd followed by the path to your Causeway folder.*

You should now be inside your Causeway folder and ready to run commands.

3. Make Causeway executable

On macOS and Linux, you may need to give your system permission to execute the Causeway binary.

Run: chmod +x cwp2p

You only need to do this once. **Windows users:** You can skip this step. Windows executables do not require the chmod command.

4. Start Causeway

From your Causeway folder, run the appropriate command.

macOS / Linux

./cwp2p

Windows (PowerShell)

.\cwp2p.exe

This starts the **Causeway daemon**, which runs the Causeway service on your device.

The terminal window where you run this command becomes your daemon window.

Keep this terminal window open while using Causeway. Closing it will stop the running process.

By default, Causeway starts using WireGuard. Other operating modes, including data channels, are available through its command-line options.

6. Wait for your NetcoreNetwork ID

Once Causeway starts successfully, your terminal should display startup information, including the Causeway version.

Wait until your **NetcoreNetwork ID** appears.

Your NetcoreNetwork ID uniquely identifies your device on the network and is used to establish connections with other Causeway peers.

Once your ID appears, Causeway is running.

Keep this terminal window open. This is your Causeway daemon.

7. Open a second terminal window

While the Causeway daemon is running, open a second terminal window or tab. Navigate into the same Causeway folder using the instructions from Step 2.

You should now have two terminal windows:

- **Terminal 1:** Runs the Causeway daemon.
- **Terminal 2:** Lets you interact with Causeway using commands.

Keep Terminal 1 running while using Terminal 2.

8. Check your Causeway status

In your second terminal window, run the status command.

macOS / Linux

```
./cwp2p status
```

Windows (PowerShell)

```
.\cwp2p.exe status
```

This lets you confirm that Causeway is running and view information about your Causeway instance.

Connect your team to NetcoreWorkspaces

Now that Causeway is running, you can use it to connect your teammates to NetcoreWorkspaces. Causeway allows devices on different networks to communicate securely, so your teammates can access the workspace even when they're not on the same Wi-Fi network.

For this setup, one device hosts NetcoreWorkspaces, and the other devices connect to it through Causeway.

Step 1: Identify your workspace host

The **workspace host** is the device running NetcoreWorkspaces.

The host should have both NetcoreWorkspaces and Causeway running.

Each teammate also needs Causeway running on their own device, but they do not need to host another instance of the application.

Step 2: Allow the workspace ports on the host

On the **host device**, find the port NetcoreWorkspaces is running on. This is the port number shown when you start the application, or in your browser's address bar after localhost:.

For example, if your browser shows http://localhost:8082, your port is 8082.

Open a second terminal window in your Causeway folder and run:

./cwp2p port add <port>

Replace <port> with the port your NetcoreWorkspaces application is using.

For example, if NetcoreWorkspaces is running on port 8082: ./cwp2p port add 8082

This allows your teammates to access your workspace through Causeway.

Step 3: Share your NetcoreNetwork ID

The workspace host should share their NetcoreNetwork ID with the teammates they want to connect.

You can find this ID in your Causeway daemon output or by running the status command.

macOS / Linux

./cwp2p status

Windows (PowerShell)

.\cwp2p.exe status

Each teammate will use this ID to add the workspace host as a peer.

Step 4: Add the workspace host as a peer

On each teammate's device, make sure Causeway is running. Follow Steps 1-8 at the beginning of this document.

In the second terminal window, add the workspace host using their NetcoreNetwork ID.

macOS / Linux

./cwp2p peer add workspace-host-name HOST_ID {port:port}

Windows (PowerShell)

.\cwp2p.exe peer add workspace-host-name HOST_ID {port:port}

Replace HOST_ID with the actual NetcoreNetwork ID shared by your workspace host. Replace workspace-host-name with whatever name you want to give to that peer and replace {port:port} with the port that the host is running Workspaces on. For the example used earlier in this document, that would be 8082:8082.

Here's what the command means:

- **workspace-host:** A friendly name for the device you're connecting to. You can choose a different name.
- **HOST_ID:** The NetcoreNetwork ID of the device hosting the workspace.
- **port:port:** Maps your local port to the host's port.

This allows the teammate's device to reach the workspace services through Causeway.

Step 5: Have the workspace host add you as a peer

Now that you've added the workspace host, they need to add you as a peer on their device.

1. Find your own **NetcoreNetwork ID** by running the Causeway status command.
2. Share your ID with the person hosting NetcoreWorkspaces.
3. The host should add your device as a peer using your NetcoreNetwork ID.

On the host's device:

macOS / Linux

```
./cwp2p peer add teammate-name TEAMMATE_ID
```

Windows (PowerShell)

```
.\cwp2p.exe peer add teammate-name TEAMMATE_ID
```

Replace `teammate-name` with a recognizable name and `TEAMMATE_ID` with the teammate's NetcoreNetwork ID.

Important: The host does not need to specify a port when adding you as a peer. Only the teammate connecting to the workspace needs to specify the host's available ports.

Step 5: Confirm your peer connection

Once you've added the workspace host, check your peer list.

macOS / Linux

./cwp2p status

Windows (PowerShell)

```
.\cwp2p.exe status
```

You should see the workspace host in your list of configured peers.

Step 6: Open NetcoreWorkspaces

Once your Causeway connection is established, open a web browser on the teammate's device.

Navigate to:

```
http://localhost:8082
```

Even though NetcoreWorkspaces is hosted on another device, Causeway allows you to access it through your own localhost.

If the connection is working correctly, NetcoreWorkspaces should load in your browser.

You're connected!

Your team can now access the same NetcoreWorkspaces instance from different devices and networks.

Keep Causeway running on each device, and keep NetcoreWorkspaces running on the host.

You're ready to collaborate!